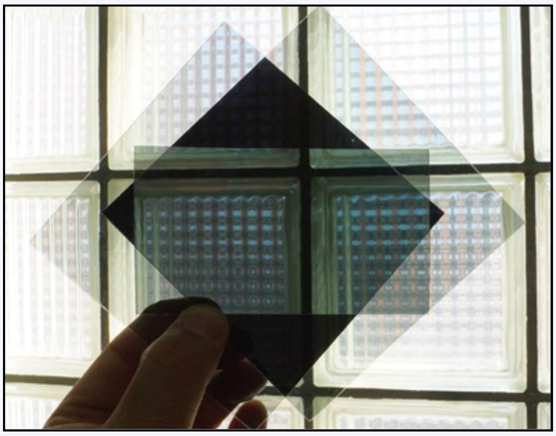
TECHNICAL DATA SHEET (FOR EXPERTS / EDUCATORS) – WORKSHOP 6 : OPTRONICS & BIREFRINGENCE OF MATERIALS

Polarizing films



Polarization Colors

<u>Background</u> : This educational workshop demonstrates the interaction between polarized light and the molecular structure of transparent polymers subjected to internal or external mechanical stresses.

<u>Objectives of the experiment</u> : to visually demonstrate the principle of 90° cross-polarization (total extinction of the light beam) and to illustrate the phenomenon of accidental birefringence using everyday injection-molded plastic objects.

I. Physical Phenomena in Focus

- *<u>Linear polarization and transmittance (T)</u>* : A polarizing film acts as a linear filter that allows only photons whose electric field oscillates along a specific axis to pass through. When two filters are stacked at 90° (orthogonal axes), the transmittance theoretically drops to $T \simeq 0$, resulting in complete light extinction and a completely black background.

- *<u>Strain-induced birefringence</u>* : Transparent injection-molded plastic objects (polymers) contain mechanical stresses and internal strains that become fixed during industrial cooling. These stresses locally alter the material's refractive index in a directional manner.

- *<u>Photoelastic analysis and color spectrum</u>* : When the plastic is inserted between the two crossed filters, the incident white light splits into two waves traveling at different speeds. Upon exiting the second filter, these waves interfere to reveal a spectrum of vivid colors (colored fringes), which is directly proportional to the intensity of the internal mechanical stresses.

II. Experimental Protocol and Technical Alignment

1. *<u>Extinction setup</u>* : Remove the protective films from both filters (dust the adhesive side with a light layer of talcum powder if necessary). Place the first filter in front of the white light, then superimpose the second filter at a 90° angle until maximum light blockage (deep black) is achieved.
2. *<u>Insertion and twisting</u>* : Slide a rigid, transparent plastic object (such as a piece of tape or a fork) between the two crossed filters. Apply a slight manual twist to the object to instantly reveal and shift the interference color fringes.

DID YOU KNOW THAT ?

Did you know ? The secret colors of invisible objects !
Why do completely transparent pieces of plastic light up with magnificent rainbows when placed between two magic filters ? It's all about internal stress ! When these objects are manufactured in the factory, they're molded very quickly, trapping invisible forces within their structure. Polarizing filters act like microscopic detectors: they force light to reveal these stresses in the form of dazzling colors. If you gently twist the object, the colors shift in real time ! It is precisely this principle, called photoelasticity, that allows civil engineers to test the strength of bridge or building models, or to detect even the tiniest microcracks in airplane windows to ensure our safety *(Academic source: IEEE Transactions on Instrumentation and Measurement).*